PROBLEM 31

Two Particles on Cone with String

Original statement · Problems.pdf, p. 19

Multi-Particle Cable Constraint

Two heavy point particles P_1 and P_2 of mass m each are connected by an inextensible, massless string of total length ℓ. Particle P_1 is constrained to move without friction over the surface of a cone of semiangle α (the particle cannot leave the surface of the cone). The string passes through an aperture at the cone vertex O, and it can slide through it without friction. Particle P_2 hangs along the cone axis, and is constrained to move along it. We define an inertial reference frame S_0. We call ξ the length of the fraction of string between O and P_1, and θ the angle formed between the Ox axis and the projection onto the Oxy plane of the $\mathbf{OP}_1$ vector. In the following, assume that the string is always tense and that the particles do not reach point O during motion.

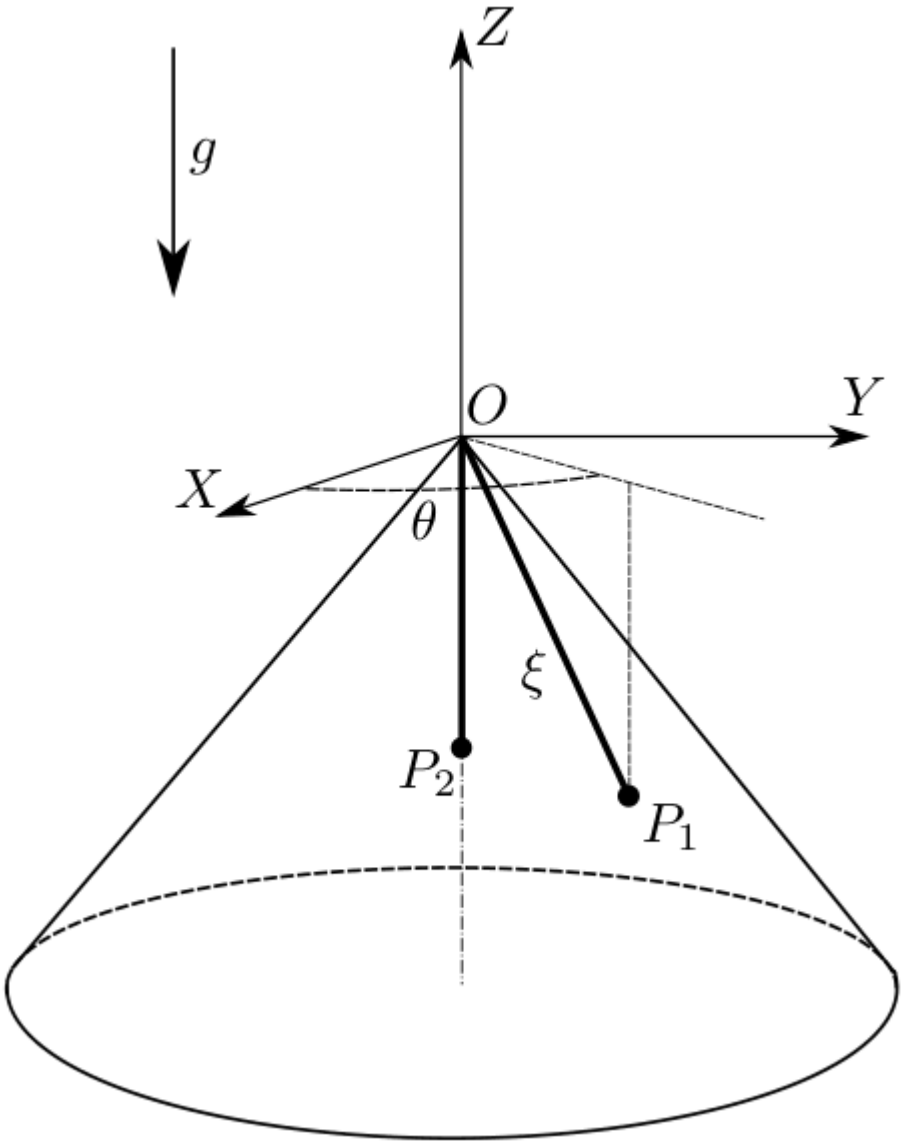


Figure: Cone with vertex O and axes X, Y, Z; P1 on the cone surface at string length ξ and angle θ, P2 hanging along the axis (Problems.pdf, p. 19).

(a) Write down the general expression of the position, velocity, and acceleration vectors of P_1 and P_2 as a function of θ, ξ and the parameters of the problem.

(b) Write down the expression of all forces that act on P_1 and P_2.

(c) Write down the differential equations that, if solved, would yield the time evolution of θ, ξ, the tension on the string, and any other reaction force in the problem.

(d) What is the mathematical condition that the function $\xi(t)$ must satisfy for the string to remain tense all the time?

(e) How fast should P_1 rotate in the θ direction to have a circular trajectory over a cone parallel? What is the value of the string tension and any other reaction force in the problem in that case? Give your answer as a function of ξ and the parameters of the problem.

(f) Taking these equations as the starting point (or based on the physical considerations of the problem) write down two first-integrals of the problem that do not depend on the string tension or any other reaction force in the problem.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two heavy particles P_1 and P_2, each of mass m, connected by an inextensible massless string of length ℓ.
- P_1 is constrained to the frictionless surface of a cone of semi-angle α with vertex at O.
- P_2 hangs vertically along the cone's axis of symmetry (Oz).
- The string passes through a frictionless hole at vertex O.
- **Configuration Degrees of Freedom (CDOFs):**
- Ambient coordinates: 6.
- Constraints: P_1 on cone (1), string inextensibility $\xi_1 + \xi_2 = \ell$ (1), P_2 on axis Oz (2).
- Total CDOFs: $6 - 4 = 2$.
- Generalized coordinates: $\xi(t)$ (distance OP_1) and azimuthal angle $\theta(t)$.
- **Hypotheses:** String is always taut ($T(t) > 0$), frictionless hole and cone surface.
- **Parameters:** $m, \ell, \alpha, g > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Cylindrical frame on the cone:
- Unit vector along cone generator from O to P_1:

$$\mathbf{e}_\xi = \sin\alpha\,\mathbf{e}_r - \cos\alpha\,\mathbf{k}$$

- Unit tangent along parallel: $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$.
- Unit outward normal to the cone:

$$\mathbf{n} = \cos\alpha\,\mathbf{e}_r + \sin\alpha\,\mathbf{k}$$

- Position of P_1:

$$\mathbf{r}_1 = \xi\mathbf{e}_\xi = \xi\sin\alpha\,\mathbf{e}_r - \xi\cos\alpha\,\mathbf{k}$$

- Position of P_2 (hanging at distance $\ell - \xi$ below O):

$$\mathbf{r}_2 = -(\ell - \xi)\mathbf{k}$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Kinematics of P_1 and P_2

1. **For P_1:**

Velocity:

$$\mathbf{v}_1 = \dot{\xi}\mathbf{e}_\xi + \xi \sin\alpha \dot{\theta}\mathbf{e}_\theta$$

Acceleration:

$$\mathbf{a}_1 = (\ddot{\xi} - \xi\dot{\theta}^2 \sin^2\alpha)\mathbf{e}_\xi + (2\dot{\xi}\dot{\theta}\sin\alpha + \xi\ddot{\theta}\sin\alpha)\mathbf{e}_\theta - \xi\dot{\theta}^2 \sin\alpha\cos\alpha\,\mathbf{n}$$

2. **For P_2:**

$$\mathbf{v}_2 = \dot{\xi}\mathbf{k}$$

$$\mathbf{a}_2 = \ddot{\xi}\mathbf{k}$$

Part (b): Forces on P_1 and P_2

1. On P_1:

- Gravity: $\mathbf{F}_{g1} = -mg\mathbf{k} = mg\cos\alpha\mathbf{e}_\xi - mg\sin\alpha\mathbf{n}$.
- String tension: $\mathbf{T}_1 = -T\mathbf{e}_\xi$ ($T > 0$).
- Normal cone reaction: $\mathbf{N} = N\mathbf{n}$.

2. On P_2:

- Gravity: $\mathbf{F}_{g2} = -mg\mathbf{k}$.
- String tension: $\mathbf{T}_2 = +T\mathbf{k}$.
- Guide reaction: $\mathbf{R}_2 \perp \mathbf{k}$.

Part (c): Equations of Motion

1. P_2 along $\mathbf{k}$:

$$m\ddot{\xi} = T - mg \implies T = m(g + \ddot{\xi})$$

2. P_1 along generator $\mathbf{e}_\xi$:

$$m(\ddot{\xi} - \xi\dot{\theta}^2 \sin^2\alpha) = mg\cos\alpha - T$$

Substitute $T = m(g+\ddot{\xi})$:

$$m(\ddot{\xi} - \xi\dot{\theta}^2 \sin^2\alpha) = mg\cos\alpha - m(g + \ddot{\xi})$$

$$2\ddot{\xi} - \xi\dot{\theta}^2\sin^2\alpha + g(1 - \cos\alpha) = 0$$

3. P_1 along parallel $\mathbf{e}_\theta$:

$$m(\xi\ddot{\theta}\sin\alpha + 2\dot{\xi}\dot{\theta}\sin\alpha) = 0 \implies \xi\ddot{\theta} + 2\dot{\xi}\dot{\theta} = 0$$

$$\frac{d}{dt}(\xi^2\dot{\theta}) = 0$$

4. P_1 along normal $\mathbf{n}$:

$$m\,\mathbf{a}_1 \cdot \mathbf{n} = N - mg\sin\alpha \implies N = mg\sin\alpha - m\xi\dot{\theta}^2\sin\alpha\cos\alpha$$

Part (d): Condition for String to Remain Taut

For the string to remain in tension ($T > 0$):

$$T = m(g + \ddot{\xi}) > 0 \implies \ddot{\xi} > -g$$

Part (e): Circular Trajectory on a Parallel

For a steady circular trajectory: $\xi = \text{constant} \implies \dot{\xi} = 0, \ddot{\xi} = 0$.

The radial equation gives:

$$-\xi\dot{\theta}^2 \sin^2\alpha + g(1 - \cos\alpha) = 0 \implies \xi\dot{\theta}^2 \sin^2\alpha = g(1 - \cos\alpha)$$

$$\dot{\theta} = \frac{1}{\sin\alpha}\sqrt{\frac{g(1 - \cos\alpha)}{\xi}}$$

Values of Tension and Normal Force:

$$T = mg$$

$$N = mg\sin\alpha - m\cos\alpha\left[\frac{g(1 - \cos\alpha)}{\sin\alpha}\right] = \frac{mg}{\sin\alpha}(\sin^2\alpha - \cos\alpha + \cos^2\alpha) = \frac{mg(1 - \cos\alpha)}{\sin\alpha}$$

Erratum: The first version had $\mathbf{a}_1 \cdot \mathbf{n} = +\xi\dot{\theta}^2 \sin\alpha\cos\alpha$ and $N = mg\sin\alpha + m\xi\dot{\theta}^2 \sin\alpha\cos\alpha$ (its own algebra in (e) did not reach the quoted result). The centripetal term points against the outward normal: $\mathbf{a}_1 \cdot \mathbf{n} = -\xi\dot{\theta}^2 \sin\alpha\cos\alpha$, hence $N = mg\sin\alpha - m\xi\dot{\theta}^2 \sin\alpha\cos\alpha$, which gives the same final value $N = mg(1 - \cos\alpha)/\sin\alpha$. Numeric check: sympy differentiation of $\mathbf{r}_1$ gives $\mathbf{a}_1 \cdot \mathbf{n} = -\frac{1}{2}\xi\dot{\theta}^2 \sin 2\alpha$ and $N/(mg) = 0.5774$ at $\alpha = 60^\circ$, equal to $(1 - \cos\alpha)/\sin\alpha$.

Part (f): Two First Integrals of Motion

1. **Azimuthal First Integral (cyclic coordinate θ):**

With $\rho = \xi \sin\alpha$ the distance of P_1 to the axis, $a_\theta = \frac{1}{\rho}\frac{d}{dt}(\rho^2\dot\theta)$. Gravity, the tension (directed to O) and the cone reaction all lie in the meridian plane of P_1, so the $\mathbf{e}_\theta$ equation reads $\frac{d}{dt}(\xi^2 \sin^2\alpha\,\dot\theta) = 0$:

$$\xi^2\dot\theta = C_1$$

2. **Mechanical Energy First Integral:**

Kinetic energy of both particles:

$$T_{total} = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m(\dot\xi^2 + \xi^2\dot\theta^2\sin^2\alpha) + \frac{1}{2}m\dot\xi^2 = m\dot\xi^2 + \frac{1}{2}m\xi^2\dot\theta^2\sin^2\alpha$$

Potential energy:

$$V_{total} = mgz_1 + mgz_2 = -mg\xi\cos\alpha - mg(\ell - \xi) = mg\xi(1-\cos\alpha) - mg\ell$$

Conserved total energy $E = T_{total} + V_{total}$:

$$2\dot\xi^2 + \xi^2\dot\theta^2\sin^2\alpha + 2g\xi(1-\cos\alpha) = C_2$$

Compare with the official solution key (Problems.pdf, Solution keys section).

PHASE 4

Physical Interpretation & Dimensional Verification

- The effective potential $V_{eff}(\xi) = \frac{mC_1^2 \sin^2 \alpha}{2\xi^2} + mg\xi(1 - \cos \alpha)$ exhibits a stable minimum corresponding to the circular parallel orbit.

Source: Problems.pdf (Page 19)